Event Driven CI/CD Pipelines

Enable Asynchronous CD without Skynet happening

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The Software Supply Chain

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The pipeline produces…

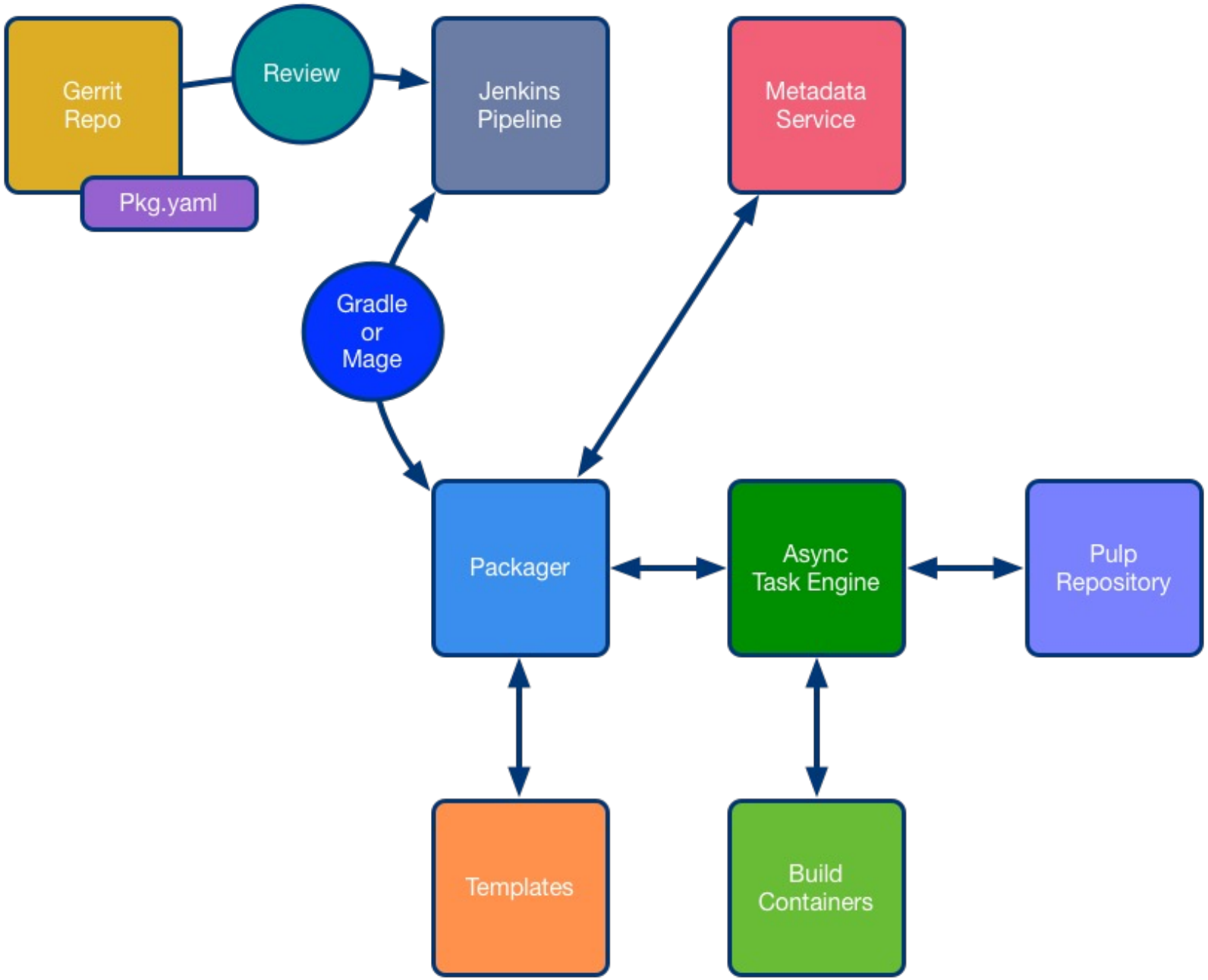
Packages
- RPMS
- DPKG
- OCI Images

Architectures
- x86_64
- aarch64
- ppc64le

Deployments
- Bare Metal
- VMs
- Kubernetes

*msi and msm packages as well

The pipeline before CI/CD Events



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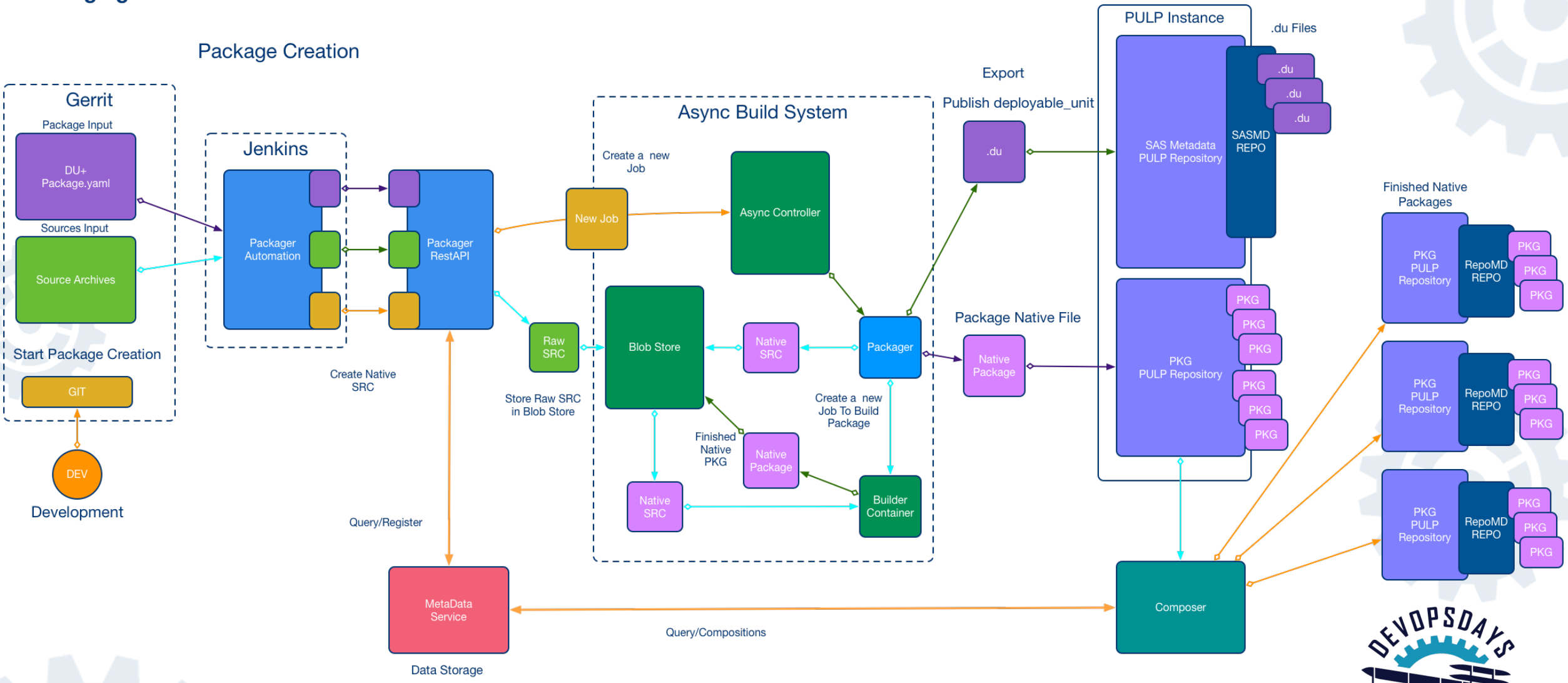
Packaging Automation



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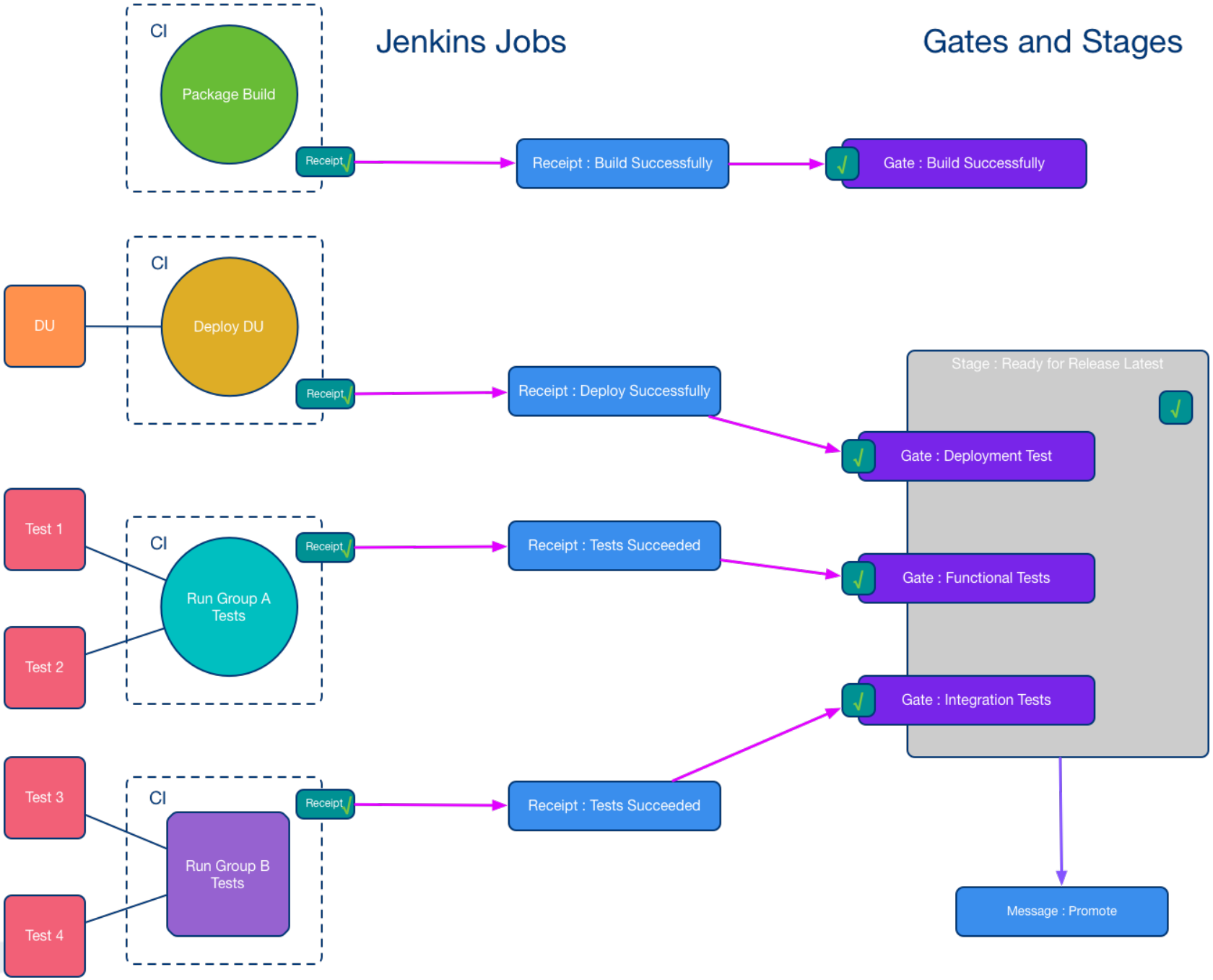
Why Event Driven CI/CD?

The Audit Trail and Gating

- Actions
- Receipts
- Gates
- Stages



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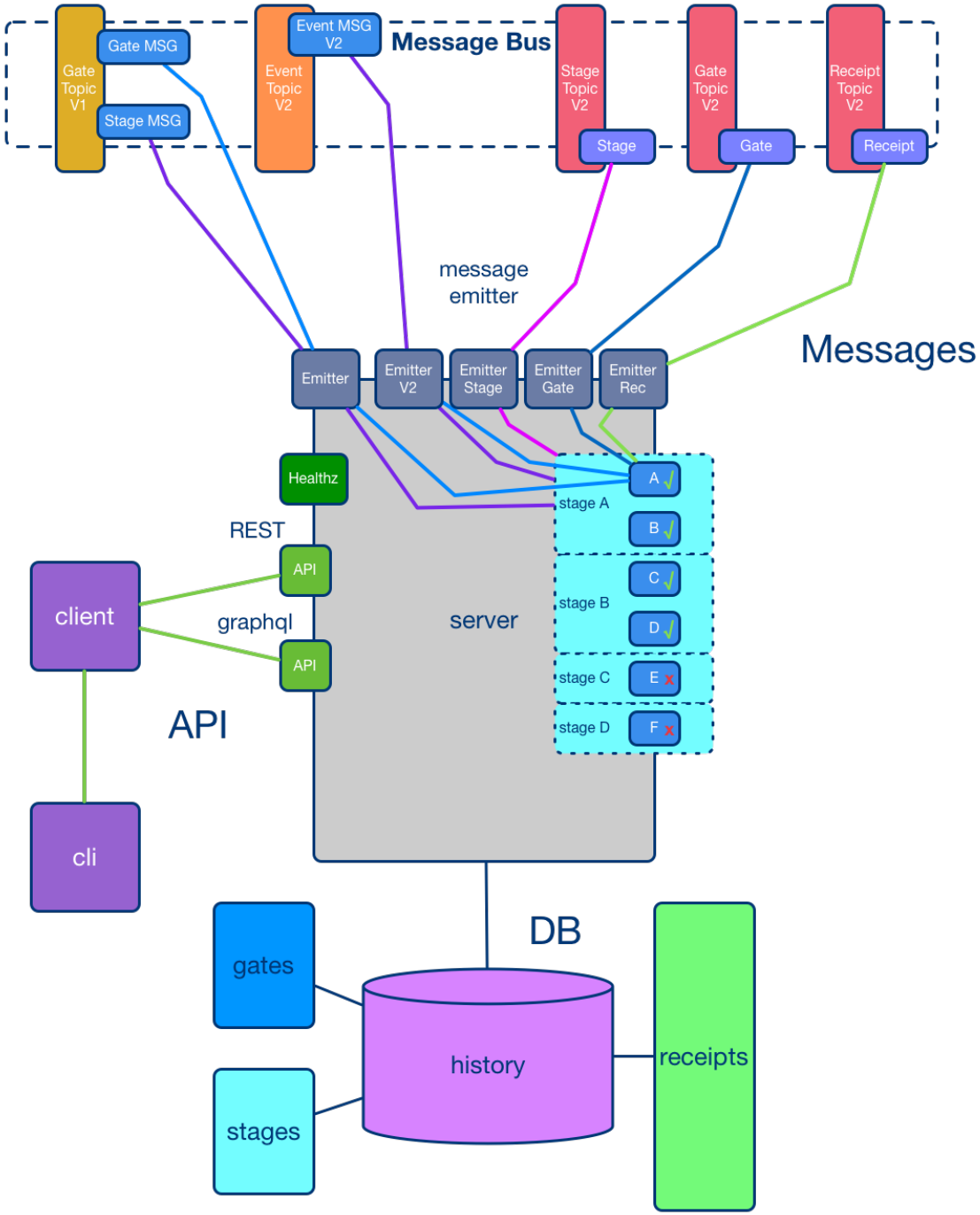

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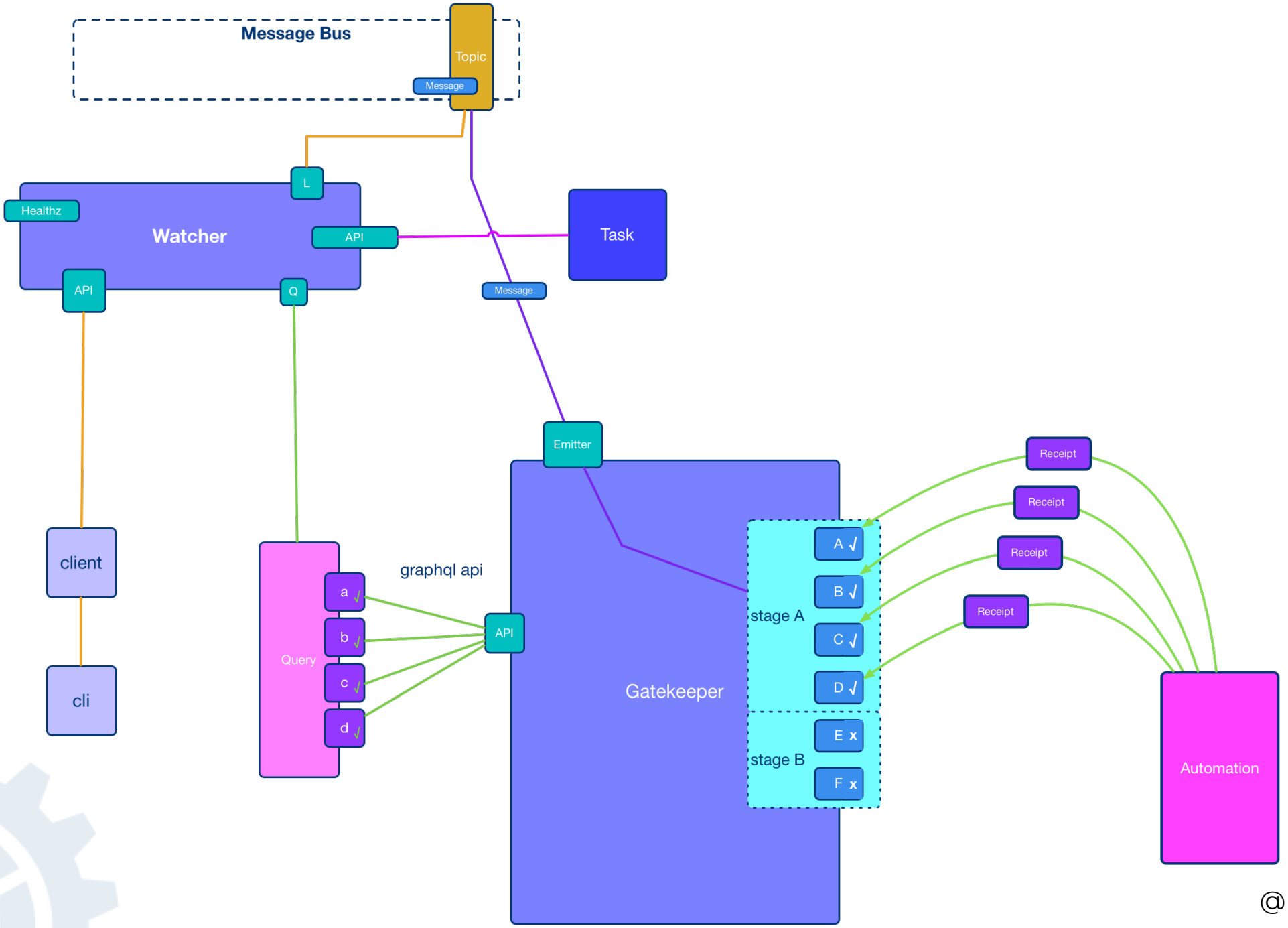

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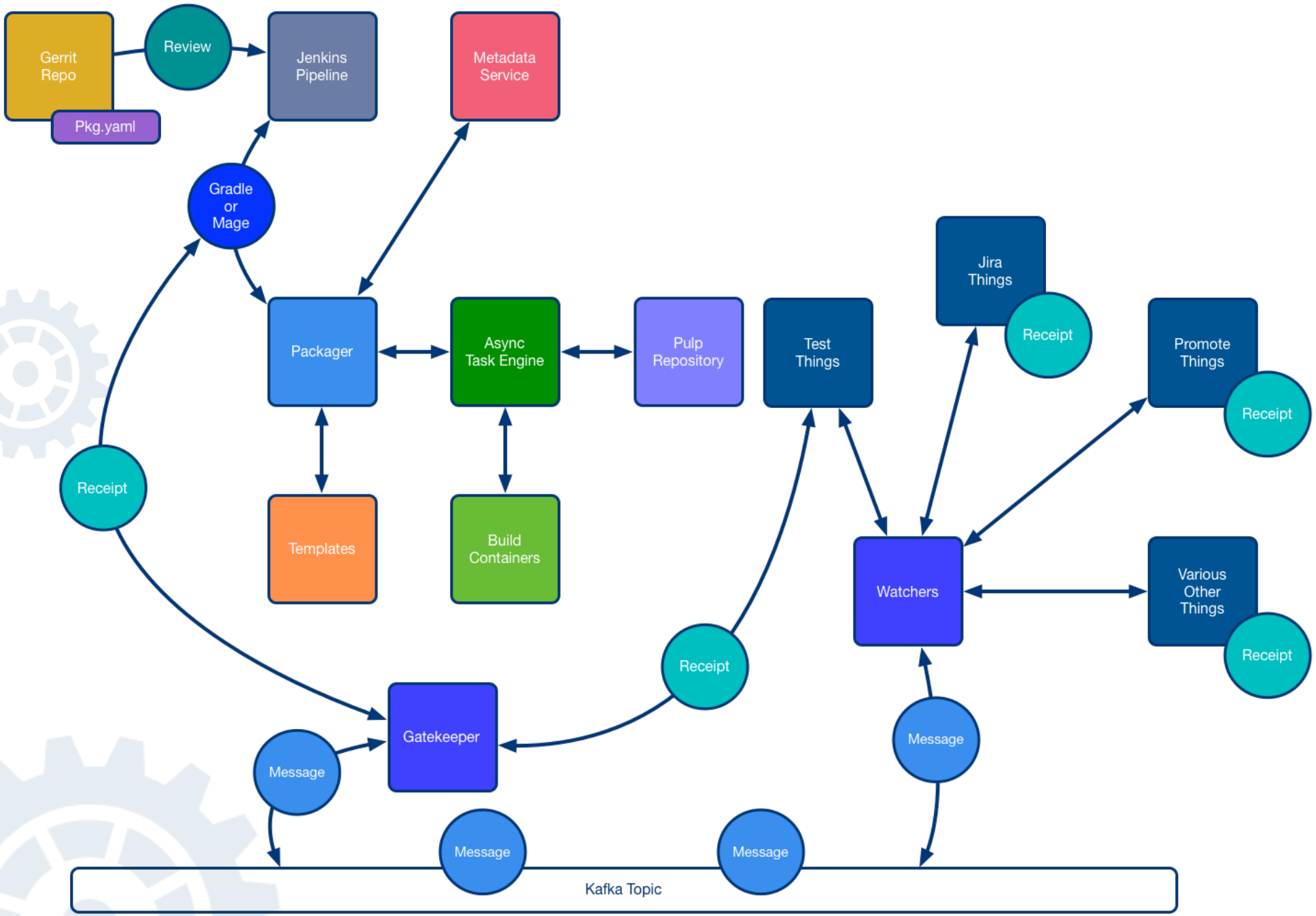
The pipeline after CI/CD Events



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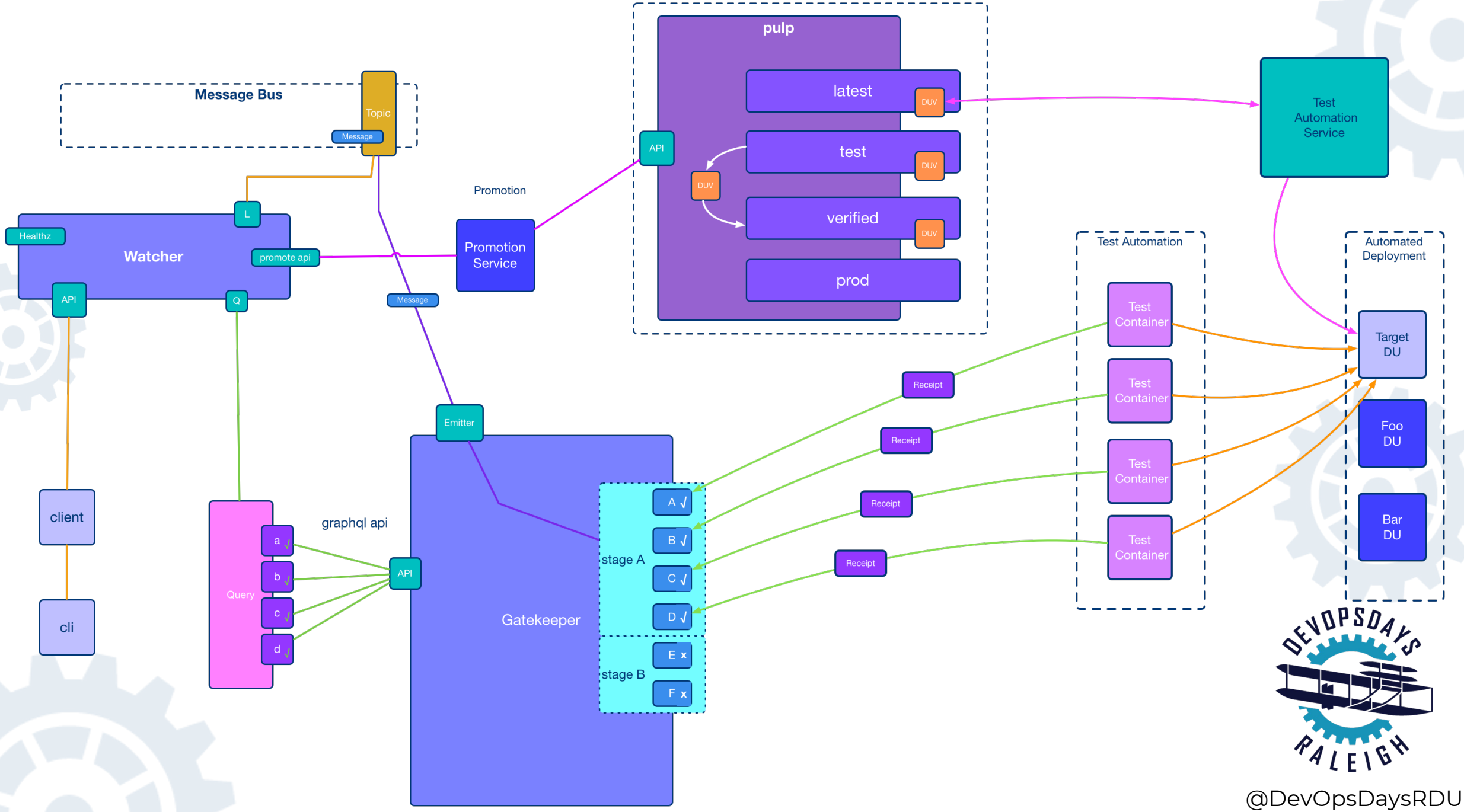
Message Bus
Topic
Message
pulp
latest
test
verified
prod
DUV
API
Test Automation Service
Promotion
Promotion Service
Healthz
Watcher
L
promote api
API
Q
Message
Emitter
client
cli
Query
a
b
c
d
graphql api
API
Gatekeeper
stage A
stage B
A √
B √
C √
D √
E x
F x
Receipt
Receipt
Receipt
Receipt
Test Automation
Test Container
Test Container
Test Container
Test Container
Automated Deployment
Target DU
Foo DU
Bar DU
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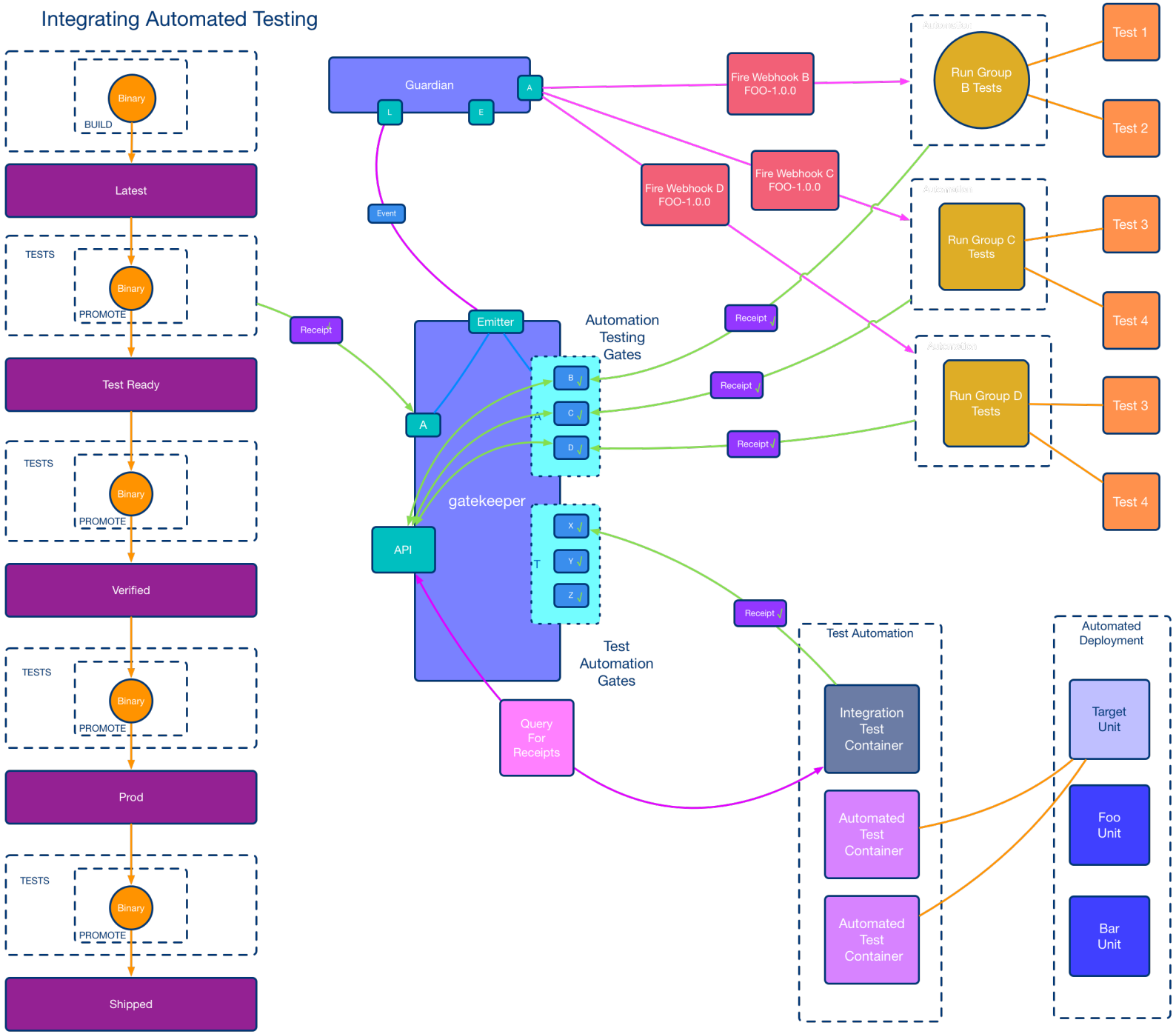
Integrating Automated Testing



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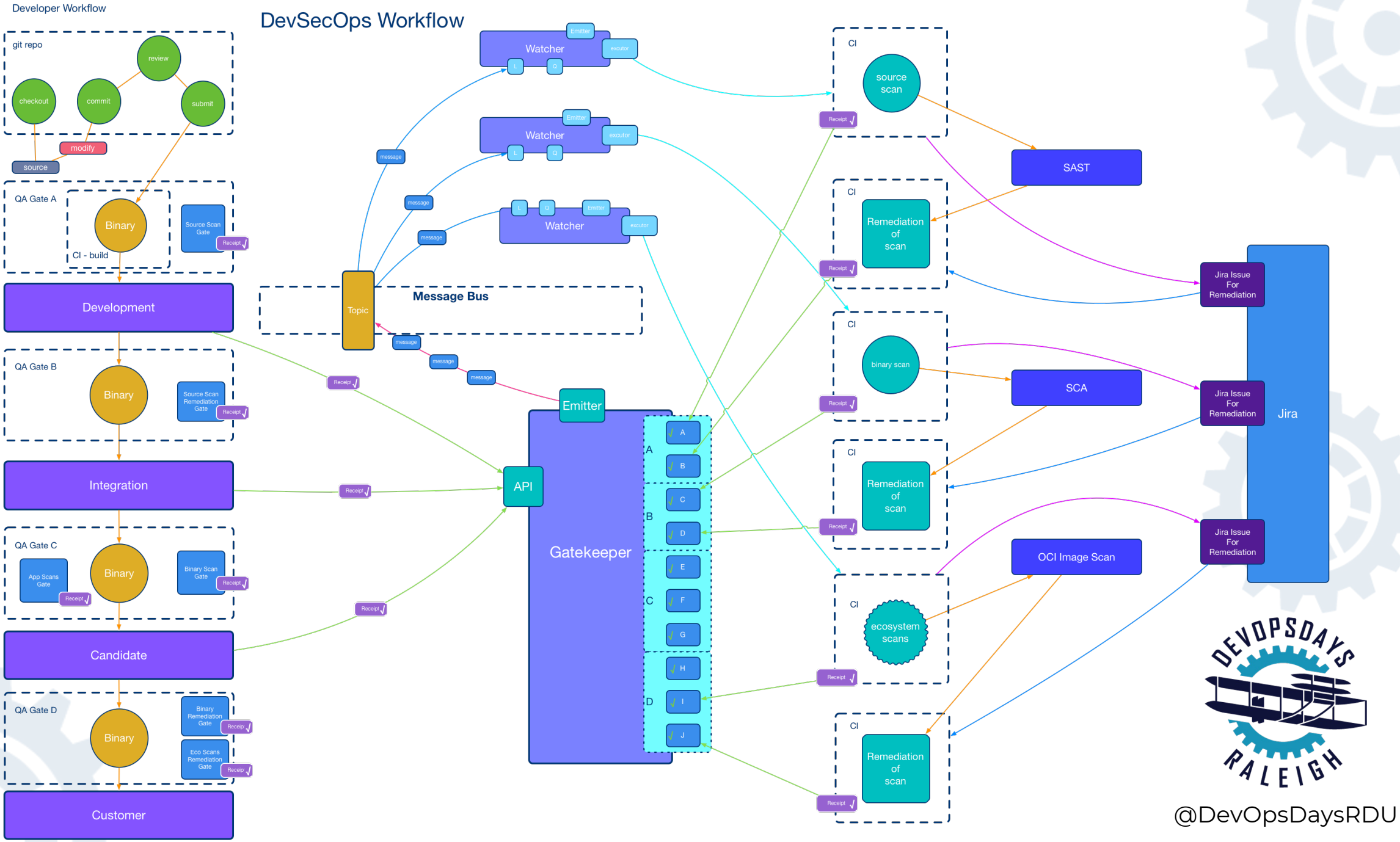
DevSecOps Workflow



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Lessons Learned

Observability

JSON Schema

Xunit Schema

Authentication (OIDC)

Event Message Size

Watcher SDK

CNAB

Had we only…

Cryptographic Signing of Receipts

RBAC Service

Observability

CD Foundation

CDFoundation
https://cd.foundation

cdevents - A common specification for Continuous Delivery events
https://cdevents.dev

SIG Events
https://github.com/cdfoundation/sig-events

SIG Software Supply Chain
https://github.com/cdfoundation/sig-software-supply-chain

- Hope is not a strategy (truth)
- Electric Sheep harmed (0)
- Receipts, Gates, Stages (done)
- Everything gets a receipt (mostly done)
- Robots promote the units (done)
- Letter to Deckard (written)
- Automatic not just Automated (mostly done)
- Reproduce actions from Receipts (not quite how I wanted but close)
- Trace the lifecycle of a binary and how it got there (done)
- Machine learning to Optimize (not yet)
- Predictive test selection (not yet)
- Deployment Risk Prediction (not yet)
- Afraid of robots (true)

I am Smitty and I am afraid of Robots

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